

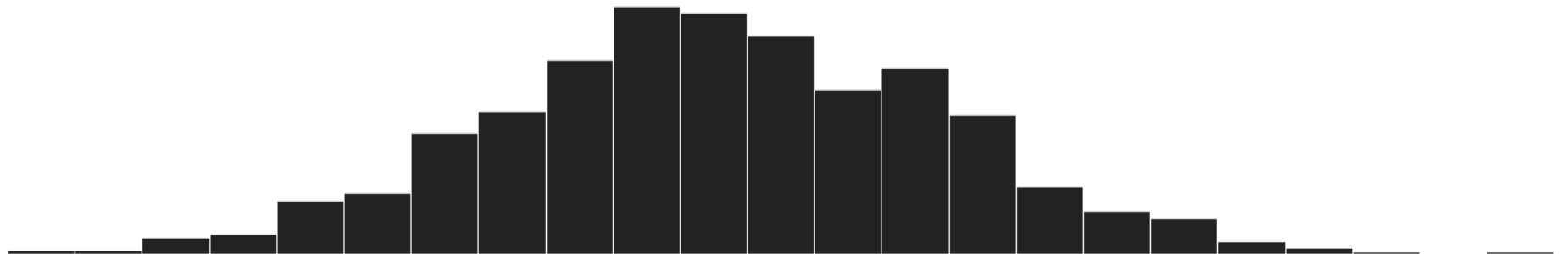
VTPEH 6105

Lecture 11

STATISTICAL FALLACIES

Amandine Gamble

amandine.gamble@cornell.edu



House Keeping

Final Exam

Wednesday May 13, 9 AM to 1 PM, LH5

Bring your laptop with

Respondus LockDown Browser installed

(link to be put on Canvas)

- Write the review of a scientific publication with a focus on data analysis and interpretation. Limited guiding questions: you should be able to identify the strengths and limitations of the publication, and assess whether you trust the conclusion(s)
- No access to AI, Google or other online resources
- Paper, hand-written cheat sheet (any shape, color, language...)
Will be checked by the teaching team

Coming Weeks

Lecture Date	Lecture & workshop	Assignment
#11 04/10	Beyond p-values: unmasking statistical fallacies Group exercises	Knowledge check #11 due Apr 17
#12 04/17	Analyzing questionnaire data <i>R how-to</i> : working with questionnaire data	Knowledge check #12 due Apr 24 Homework: Review due Apr 23
#13 04/24	Introduction to high-dimensional data analysis and machine learning Mini journal club	Knowledge check #13 due May 1 Homework: Review due Apr 30
#14 05/01	Introduction to mechanistic modeling <i>R how-to</i> : tailoring existing models	
05/13	<i>Exam</i>	<i>Review</i>

Getting Ready for the Final Exam

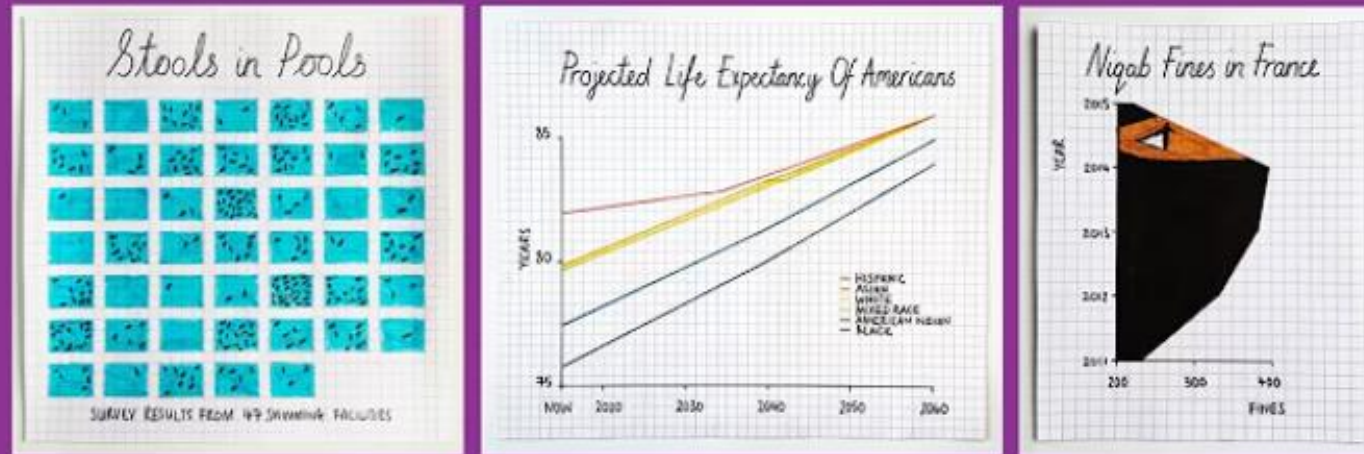
- Use the two coming homework as practice
 - Try to complete without AI, Google or other online resources
 - Add any concept you need to check to your cheat sheet
- Prepare your cheat sheet

Suggestions:

- Which test for what
- Assumptions for each test
- Best practice for data reporting and analysis
- Homework guiding questions

Refresher

How to Spot Bad Statistic



11:46

How to Assess Data Analysis in a Publication?

- The data collection and analysis protocols are clear
 - Ideally, should contain enough information to replicate the study

How to Assess Data Analysis in a Publication?

- The data collection and analysis protocols are clear
- The data is represented honestly and thoroughly
 - Ideally with visuals showing **all the data**

Information

Histogram

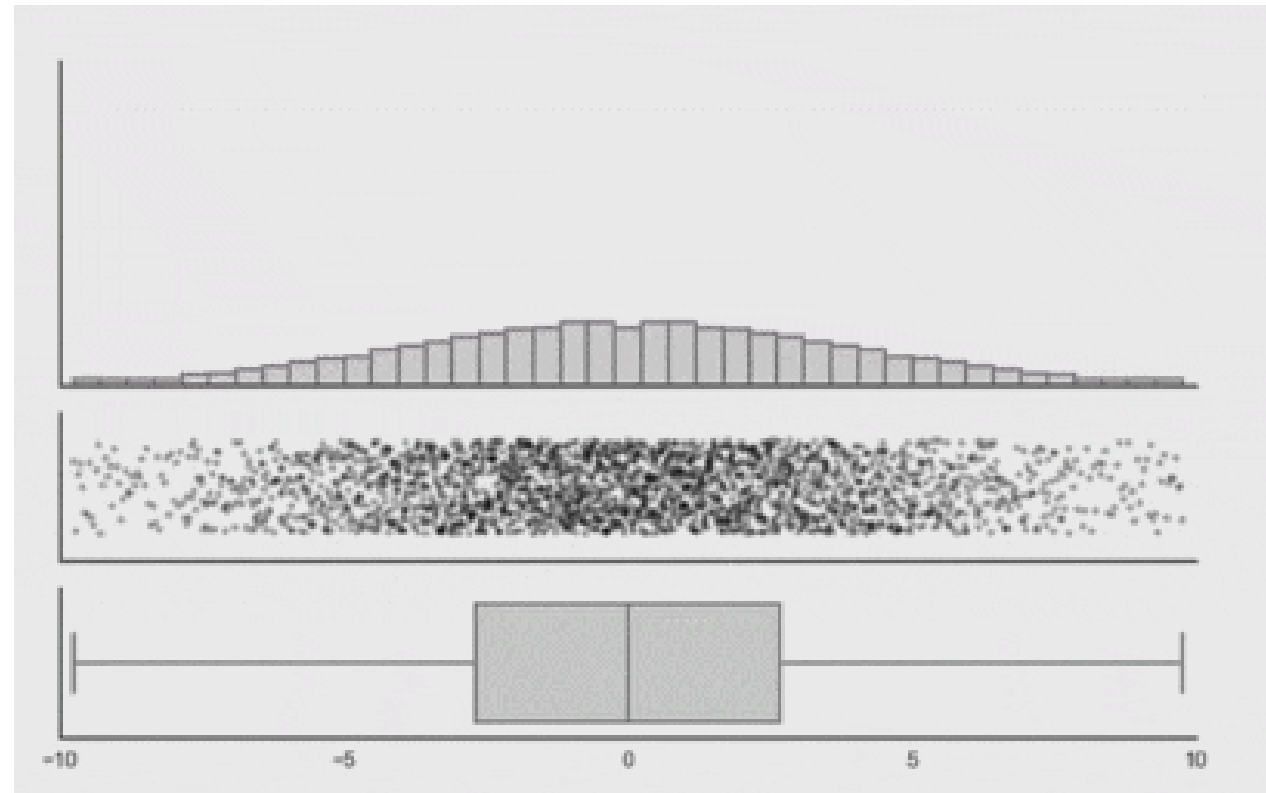
+++

Jittered
scatter plot

+++

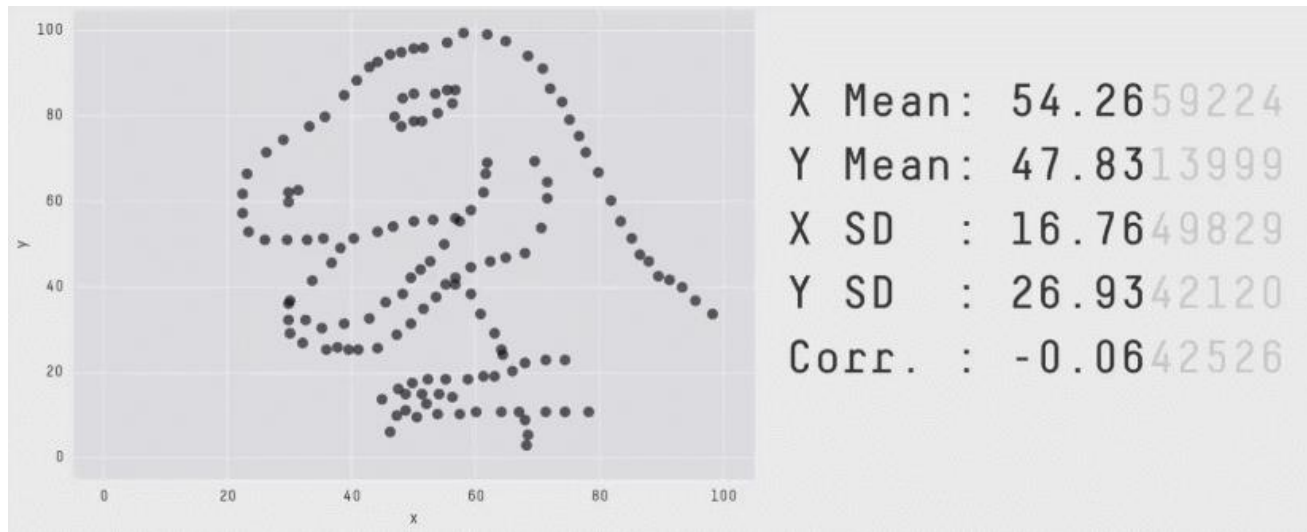
Boxplot
Summary statistics

-



How to Assess Data Analysis in a Publication?

- The data collection and analysis protocols are clear
- The data is represented honestly and thoroughly
 - Ideally with visuals showing **all the data**
 - Summaries can be used but with caution, and always include measures of both...
Central tendency **and dispersion**



How to Assess Data Analysis in a Publication?

- The data collection and analysis protocols are clear
- The data is represented honestly and thoroughly
- The appropriate method is picked to analyze the data to answer the question
 - Appropriate approach is picked considering the type and number of variables

Hypothesis Testing – Which Statistical Test?

Context	Parametric Approach	Non-Parametric Approach
Continuous variable ~ categorical variable		
Continuous variable ~ continuous variable		
Categorical variable ~ categorical variable		

Predictions – Which Model?

Context	Parametric Approach	Non-Parametric Approach
Continuous outcome variable		
Binary outcome variable		

Hypothesis Testing – Which Statistical Test?

Context	Parametric Approach	Non-Parametric Approach
Continuous variable ~ categorical variable		
Comparison of 2 means	T-test	Wilcoxon rank test
Comparison of > 2 means	ANOVA	Kruskal-Wallis rank test
Continuous variable ~ continuous variable		
Correlation between 2 continuous variables	Pearson correlation	Spearman rank test
Categorical variable ~ categorical variable		
Comparison of prevalences	Chi2, Fisher exact test	

Predictions – Which Model?

Context	Parametric Approach	Non-Parametric Approach
Continuous outcome variable		
Comparison of 2 means	Linear regression	NA
Binary outcome variable		
Correlation between 2 continuous variables	Logistic regression	NA

How to Assess Data Analysis in a Publication?

- The data collection and analysis protocols are clear
- The data is represented honestly and thoroughly
- The appropriate method is picked to analyze the data to answer the question
 - Appropriate approach is picked considering the type and number of variables
 - The approach assumptions are verified, and the reader knows it

Assumptions for Parametric Approaches

- Independence of observations
 - Study design
- Normality
 - Within each group (t-test, ANOVA, Pearson correlation...)
 - Of the residuals (regression models)
 - Histograms
 - Scatter plots
 - Shapiro-Wilk tests
- Homoscedasticity
 - Histograms
 - Scatter plots
- For regression models: functional shape
 - Linear regression: linear
 - Logistic regression: sigmoid
 - Scatter plots
- For multivariate models: absence of collinearity
 - Scatter plots
 - Linear regressions
 - Variance inflation factor

How to Assess Data Analysis in a Publication?

- The data collection and analysis protocols are clear
- The data is represented honestly and thoroughly
- The appropriate method is picked to analyze the data to answer the question
 - Appropriate approach is picked considering the type and number of variables
 - The approach assumptions are verified, and the reader knows it
 - Accounts for multiple testing if relevant
 - Parsimonious models are used for inference

Hypothesis Testing – Which Statistical Test? When?

Context	Parametric Approach	Non-Parametric Approach
Continuous variable ~ categorical variable		
Comparison of 2 means	T-test	Wilcoxon rank test
Comparison of > 2 means	ANOVA	Kruskal-Wallis rank test
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Predictions – Which Model? When?

Context	Parametric Approach	Non-Parametric Approach
Continuous outcome variable		
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Correlation between 2 continuous variables	Logistic regression	NA

How to Assess Data Analysis in a Publication?

- The data collection and analysis protocols are clear
- The data is represented honestly and thoroughly
- The appropriate method is picked to analyze the data to answer the question
- The analysis outputs are represented honestly and thoroughly
 - Statistical significance?
 - Amplitude?
 - Predictive power / explained variance?

How to Assess Data Analysis in a Publication?

- The data collection and analysis protocols are clear
- The data is represented honestly and thoroughly
- The appropriate method is picked to analyze the data to answer the question
- The analysis outputs are represented honestly and thoroughly
 - Statistical significance? p-value
 - Amplitude? Effect size with confidence interval (or a plot or summary statistics)
 - Predictive power / explained variance? Coefficient of determination (or a plot)

How to Assess Data Analysis in a Publication?

- The data collection and analysis protocols are clear
- The data is represented honestly and thoroughly
- The appropriate method is picked to analyze the data to answer the question
- The analysis outputs are represented honestly and thoroughly

In other words

- Can I replicate the study?
- Can I see the data?
- Would I have picked the same approach to analyze the data?
- Do the results support the conclusions?

How to Write a Good Review?

- Identify strengths
- Identify weaknesses: points that are wrong or unclear
 - + Suggest solutions

Examples: Suaárez-González et al. 2025

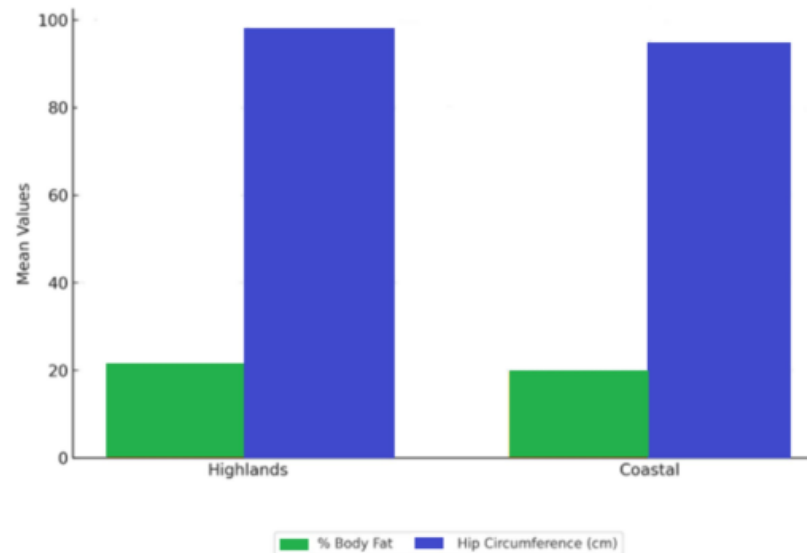


FIGURE 1
Body composition by birthplace.

How to Write a Good Review?

- Identify strengths
- Identify weaknesses: points that are wrong or unclear
 - + Suggest solutions

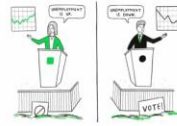
Examples: Kirchengast & Marosi, 2008

“Girls reported a significantly higher frequency of thinking concerning adequate nutrition than boys ($\chi^2 = 14.0$ $p < 0.01$).”

Statistical Fallacies

And Statistical “Don’t”

Data Fallacies to Avoid



Cherry Picking

Selecting results that fit your claim and excluding those that don't.



Data Dredging

Repeatedly testing new hypotheses against the same set of data, failing to acknowledge that most correlations will be the result of chance.



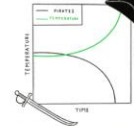
Survivorship Bias

Drawing conclusions from an incomplete set of data, because that data has 'survived' some selection criteria.



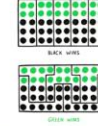
Cobra Effect

Setting an incentive that accidentally produces the opposite result to the one intended. Also known as a Perverse Incentive.



False Causality

Falsely assuming when two events appear related that one must have caused the other.



Gerrymandering

Manipulating the geographical boundaries used to group data in order to change the result.



Sampling Bias

Drawing conclusions from a set of data that isn't representative of the population you're trying to understand.



Gambler's Fallacy

Mistakenly believing that because something has happened more frequently than usual, it's now less likely to happen in future (and vice versa).



Hawthorne Effect

The act of monitoring someone can affect their behaviour, leading to spurious findings. Also known as the Observer Effect.



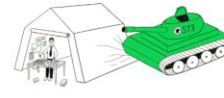
Regression Towards the Mean

When something happens that's unusually good or bad, it will revert back towards the average over time.

APPLICATION SUCCESS RATE			
	MALE	FEMALE	
GROUP 1	14% (148 of 1050)	10% (178 of 1800)	
GROUP 2	50% (448 of 890)	51% (181 of 350)	
TOTAL	28% (148 of 1050)	19% (178 of 1800)	??

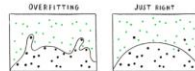
Simpson's Paradox

When a trend appears in different subsets of data but disappears or reverses when the groups are combined.



McNamara Fallacy

Relying solely on metrics in complex situations and losing sight of the bigger picture.



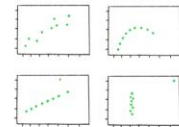
Overfitting

Creating a model that's overly tailored to the data you have and not representative of the general trend.



Publication Bias

Interesting research findings are more likely to be published, distorting our impression of reality.

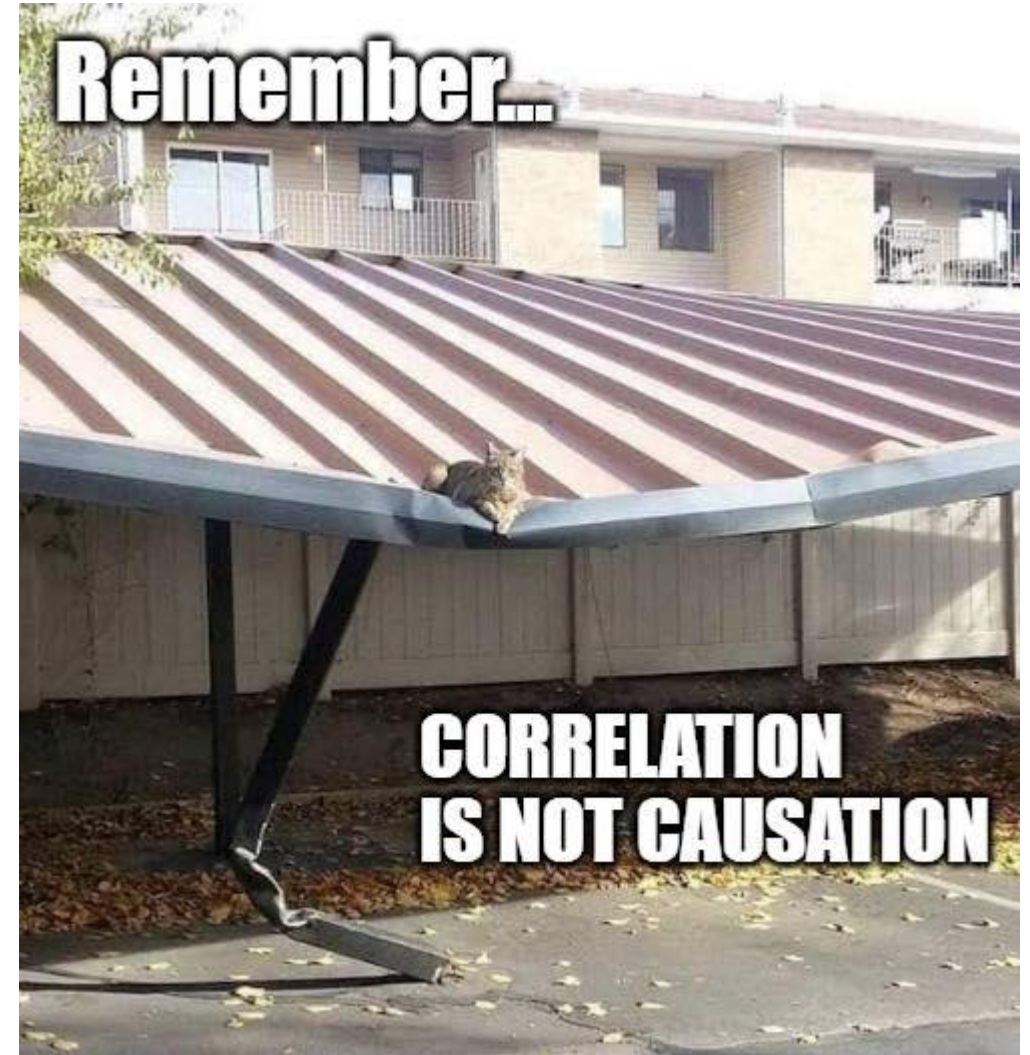
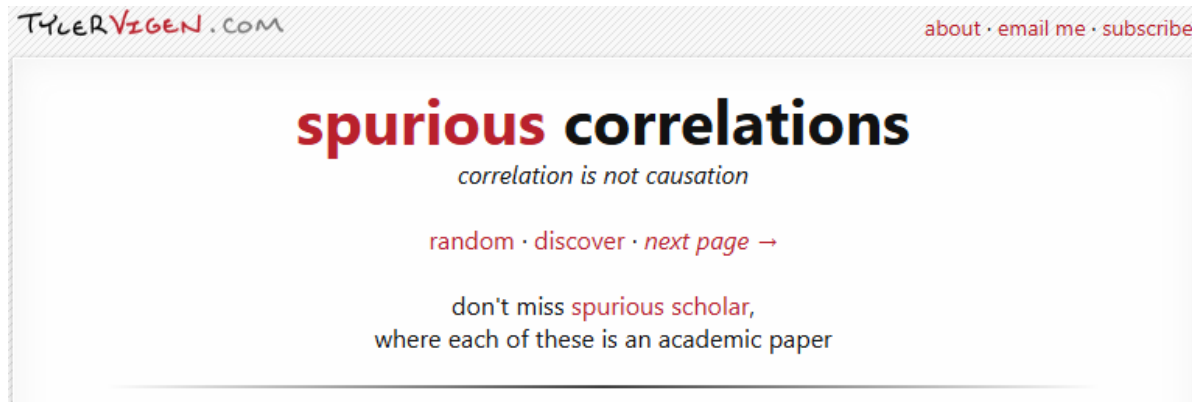


Danger of Summary Metrics

Only looking at summary metrics and missing big differences in the raw data.

False Causality

- Gold standard
 - Experimental approaches: causation
 - Observational approaches: correlation



False Causality

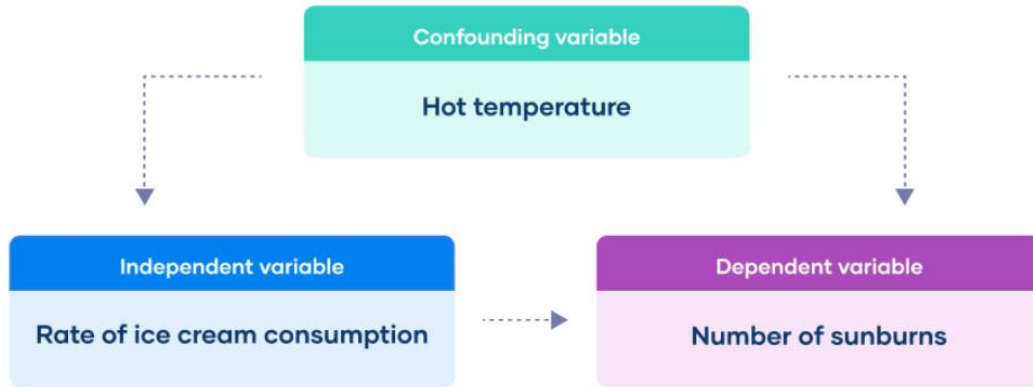
Solution?

- If correlative study: talk about correlation
- If want to talk about causality: run an experiment

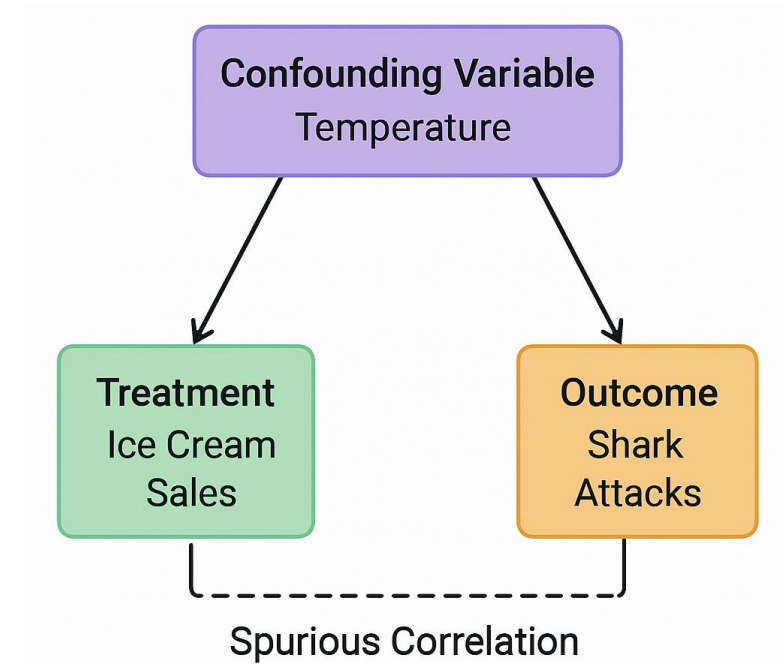
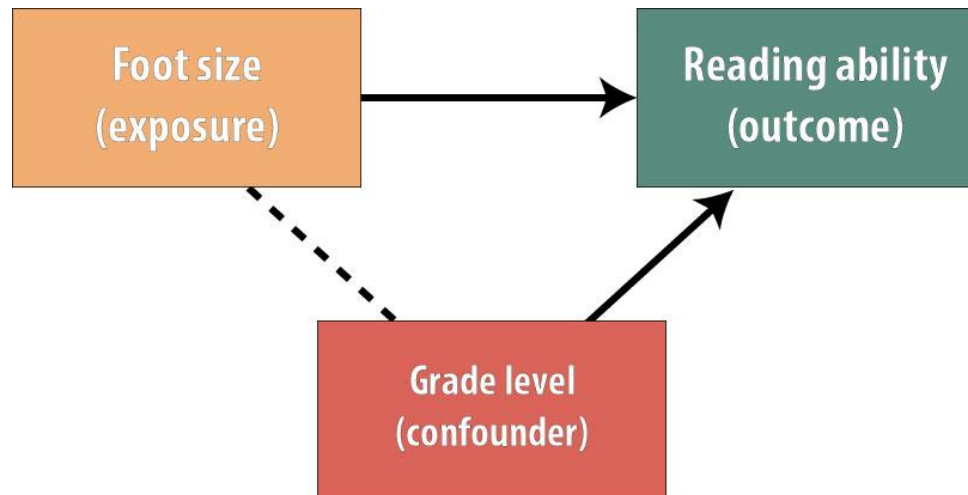
Or accumulate a lot of observational evidence, and some theory on the mechanisms

Cofounding Variables

Conceptual model with a confounding variable



 Scribbr



Cofounding Variables



Prenatal paracetamol exposure and child neurodevelopment: a systematic review and meta-analysis



Francesco D'Antonio*, Maria Elena Flacco*, Lorenza Della Valle, Smriti Prasad, Lamberto Manzoli, Athina Samara, Asma Khalil

Summary

Lancet Obstet Gynaecol Womens Health 2026; 2: e190-98

Published Online

January 16, 2026

[https://doi.org/10.1016/S3050-5038\(25\)00211-0](https://doi.org/10.1016/S3050-5038(25)00211-0)

Background Concerns have emerged about the impact of paracetamol use in pregnancy on child neurodevelopment, particularly in relation to autism spectrum disorder. We aimed to synthesise available evidence to investigate associations between prenatal paracetamol exposure and autism spectrum disorder, attention-deficit hyperactivity disorder (ADHD), and intellectual disability.

“The study clarifies that previously reported associations in conventional observational studies are likely to reflect residual confounding from maternal illness, fever, genetic susceptibility, or environmental factors rather than a causal effect of paracetamol.”

Cofounding Variables

Solution?

- Adjust for confounding variables...
- Including them in a multivariate model
- Adjustment and standardization methods
- Use paired designs

Simpson's Paradox

- Simpson's paradox is a phenomenon in probability and statistics in which a trend appears in several groups of data but disappears or reverses when the groups are combined
- Here data are combined across age classes, which were not receiving vaccination at the same rate



Dr Anthony Hinton
@TonyHinton2016

Thanks to [@abirballan](#)

orange line= vaccinated 10-59 y old

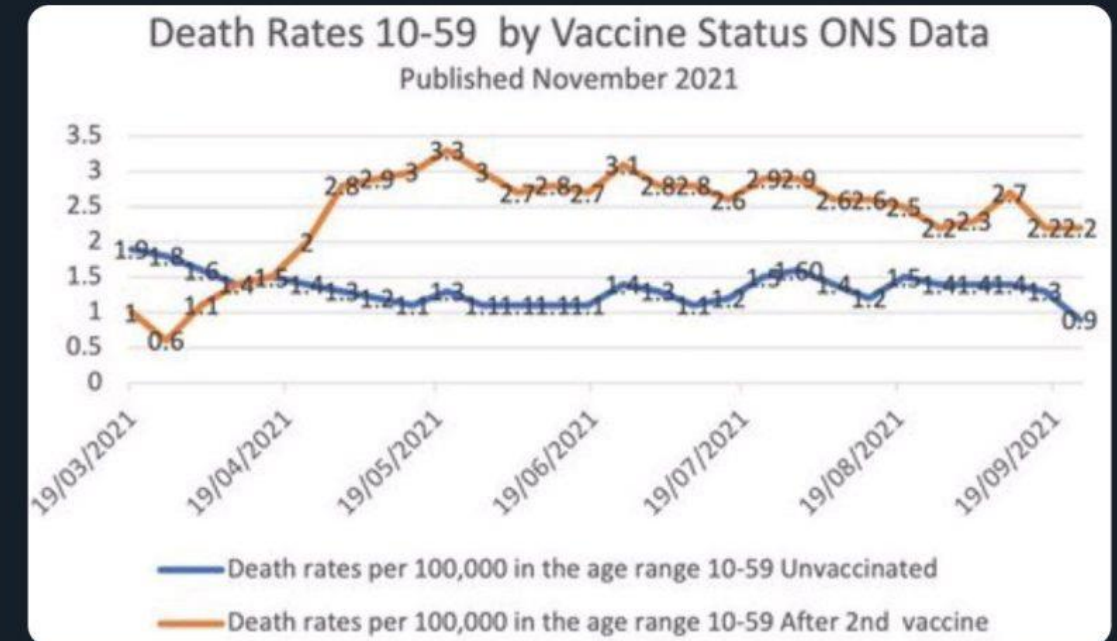
blue line= unvaccinated 10-59 y old

Orange line is ABOVE blue line

=> MORE deaths in vaccinated than unvaccinated?

Is the right policy to vaccinate low-risk populations?

ANY explanation [@CMO_England](#) [@sajidjavid](#) [@ONS](#) ???



08:58 · 20/11/2021 · [Twitter for iPad](#)

1,109 Retweets 148 Quote Tweets 1,706 Likes

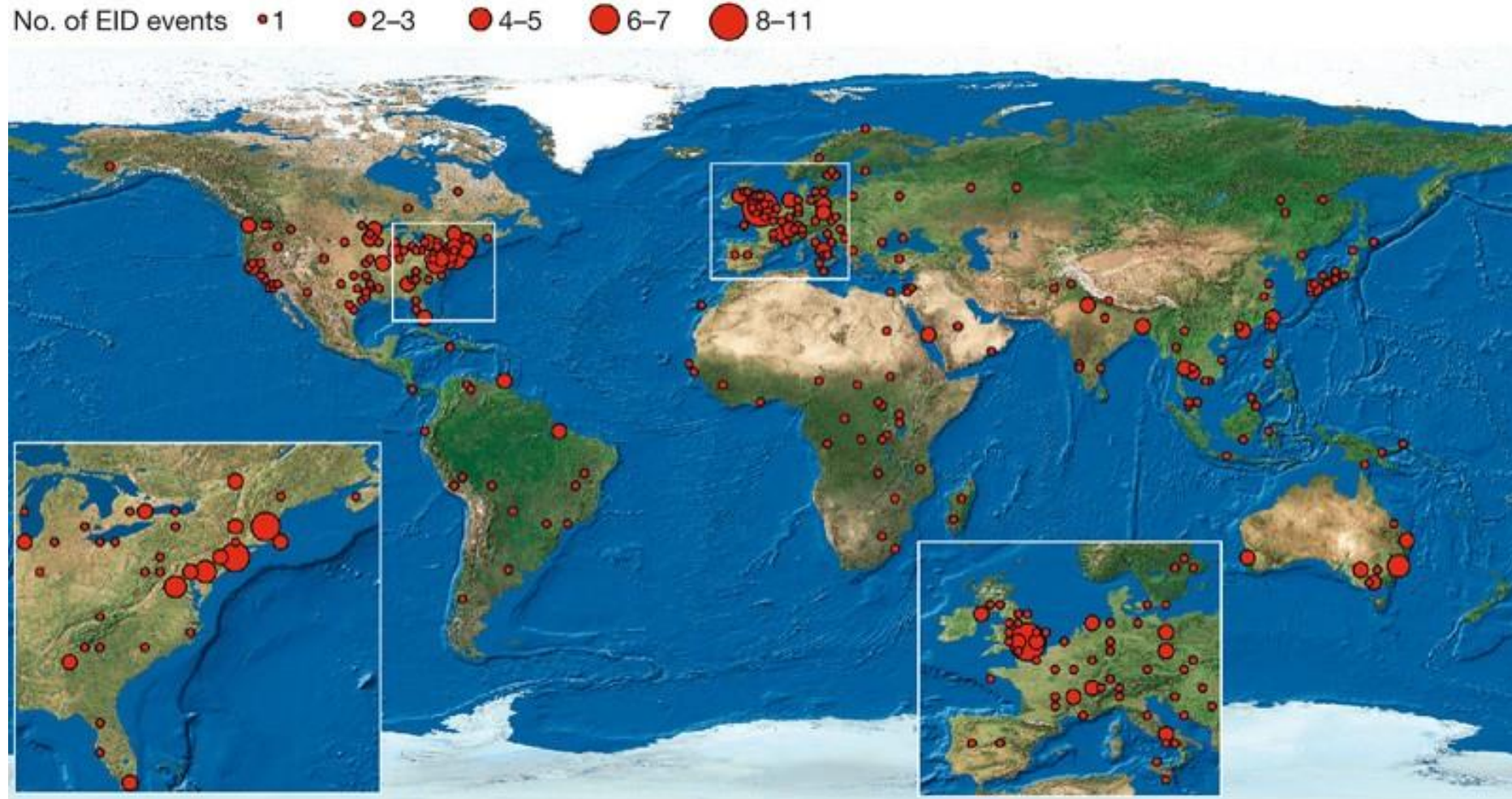
Simpson's Paradox

Solution?

- Adjust for confounding variables...
- Including them in a multivariate model
- Adjustment and standardization methods
- Use paired designs

Observation Bias

- Number of emerging infectious disease events



Observation Bias

Solution?

- Adjust for confounding variables... In particular observation effort!

More about Confounders

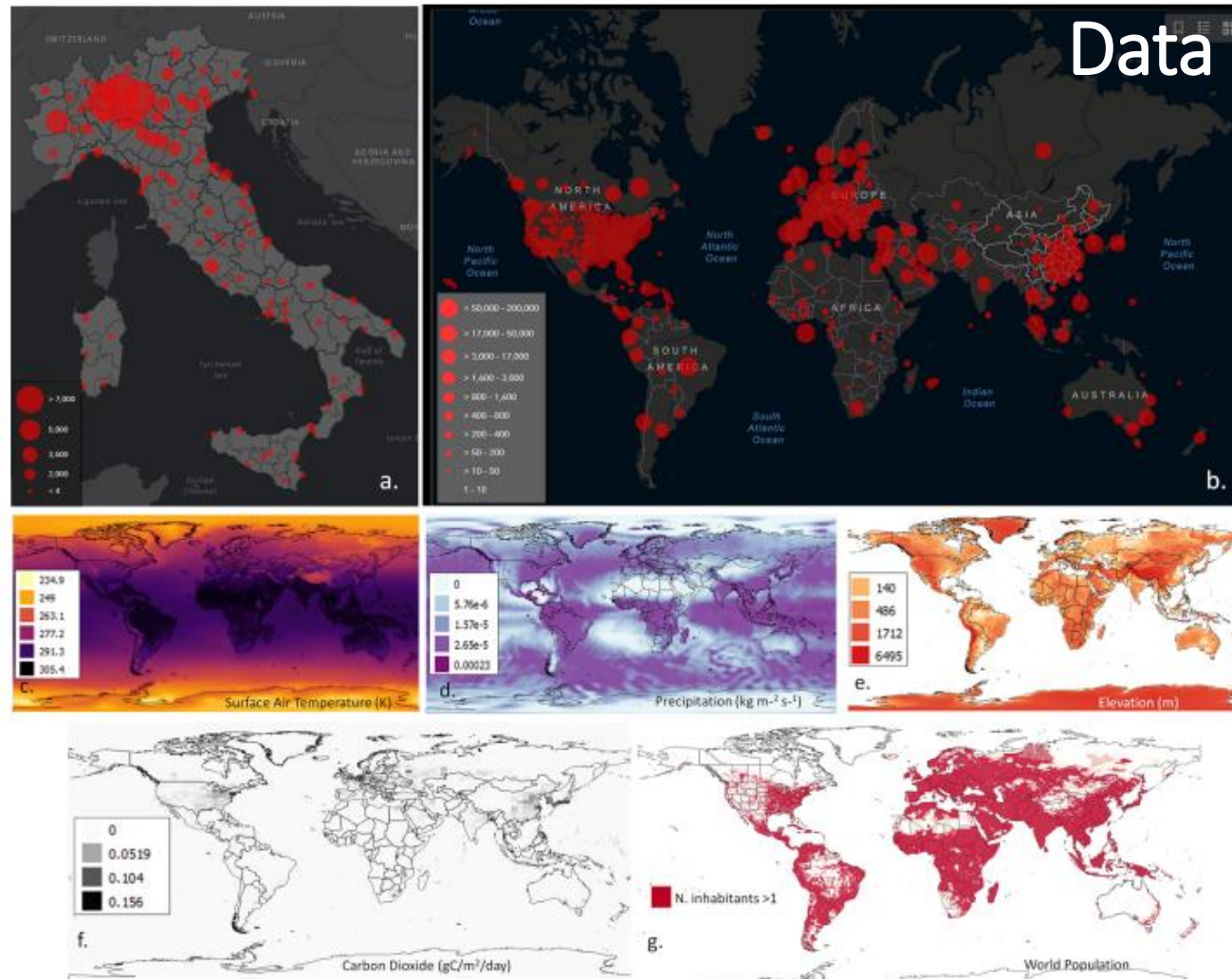
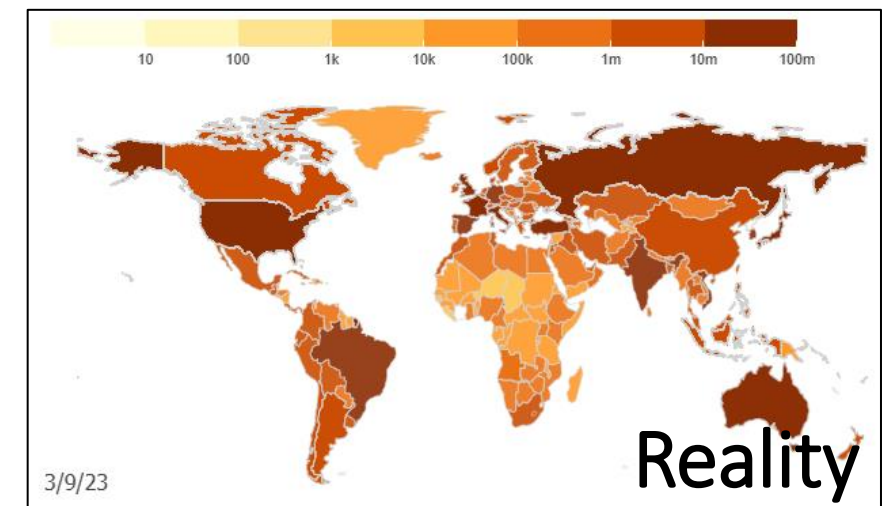
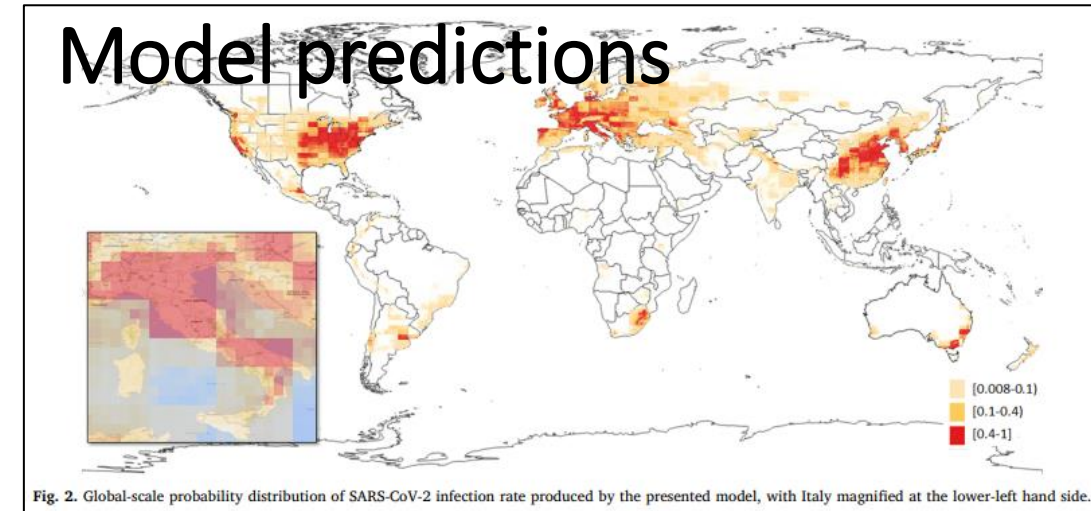


Fig. 1. Visual comparison of the global-scale data used in the presented model: (a) number of infections in Italian provinces (31 March 2020), (b) global infections (31 March 2020), (c) surface air temperature, (d) precipitation, (e) elevation, (f) carbon dioxide, (g) World population.



P-hacking

- Analyzing the data with the purpose / until a $p < 0.05$ is found
- Can be reached through...
 - Multiple testing in different forms
 - Data dredging/mining approaches
 - Subgroup analyses
 - Flexible data exclusion (e.g., being too generous on what an “outlier” is)
 - Post-hoc changes of significance thresholds
 - ...

PERSPECTIVE

The Extent and Consequences of P-Hacking in Science

Megan L. Head^{1*}, Luke Holman¹, Rob Lanfear^{1,2}, Andrew T. Kahn¹, Michael D. Jennions¹

¹ Division of Evolution, Ecology and Genetics, Research School of Biology, Australian National University, Acton, Canberra, Australia, ² Department of Biological Sciences, Faculty of Science, Macquarie University, North Ryde, New South Wales, Australia

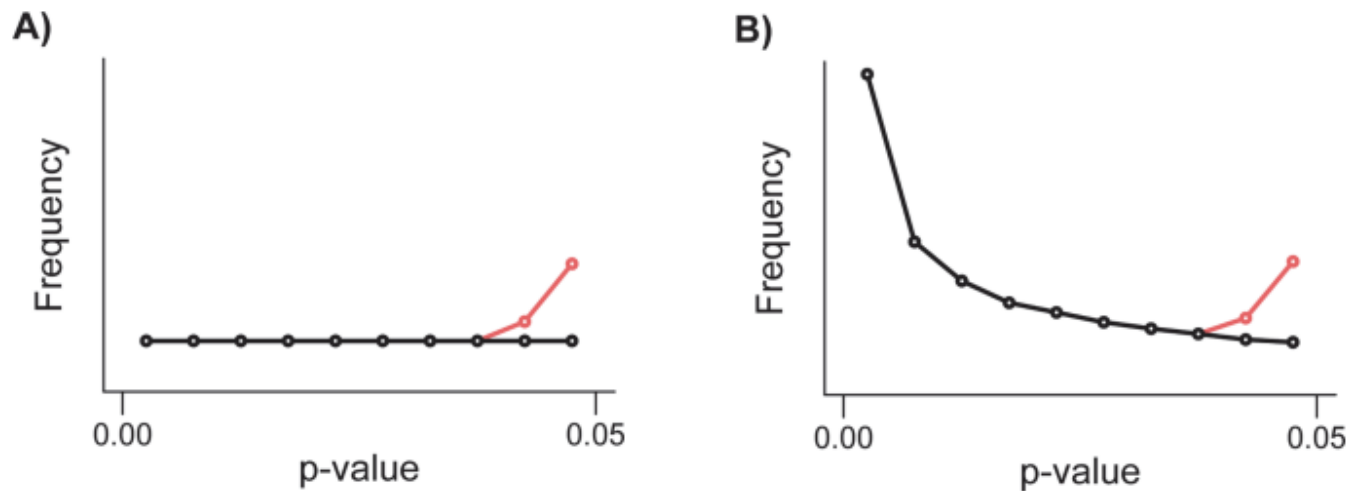


Fig 2. The effect of p-hacking on the distribution of p-values in the range of significance. A) Black line shows distribution of p -values when there is no evidential value and the red line shows how p-hacking influences this distribution. B) Black line shows distribution of p -values when there is evidential value and the red line shows how p-hacking influences this distribution. Tests for p-hacking often compare the number of p -values in two adjacent bins just below 0.05.

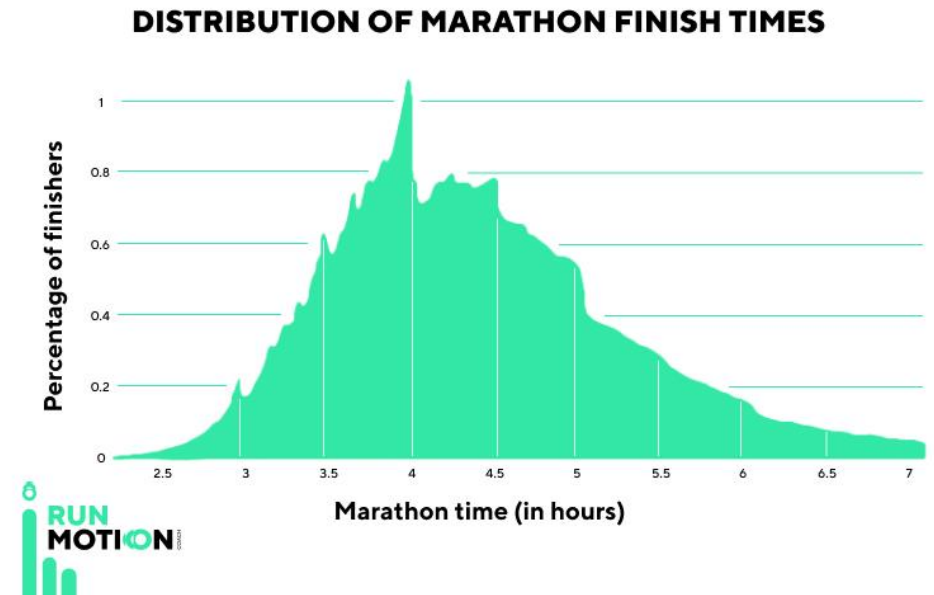


Table 1. Tests for evidential value and p-hacking across disciplines, using p -values obtained from the Results section.

Discipline	Number of p -values between 0 and 0.025	Number of p -values between 0.025 and 0.05	Binomial test for evidential value	Number of p -values between 0.04 and 0.045	Number of p -values between 0.045 and 0.05	Binomial test for p-hacking
Agricultural and veterinary sciences	375	125	<0.001	10	16	0.163
Biological sciences	11,074	3,562	<0.001	350	423	0.005
Chemical sciences	380	110	<0.001	14	17	0.360
Earth sciences	76	25	<0.001	0	4	0.063
Education	280	101	<0.001	9	8	0.685
Engineering	471	183	<0.001	16	12	0.828
Environmental sciences	657	190	<0.001	10	19	0.068
Information and computing sciences	790	266	<0.001	20	30	0.101
Mathematical sciences	72	22	<0.001	3	0	1.000
Medical and health sciences	45,460	16,537	<0.001	1,477	1,785	<0.001
Multidisciplinary	21,209	6,793	<0.001	638	750	0.001
Psychology and cognitive sciences	1,355	487	<0.001	29	50	0.012
Studies in human society	139	45	<0.001	8	3	0.967
Technology	94	37	<0.001	3	3	0.656

Number of p -values in each bin is the mean number based on 1,000 bootstraps of one p -value per Results section, rounded to the nearest whole number. Disciplines ($n = 8$) for which we found fewer than 50 p -values below 0.05 in the Results section were excluded.

PERSPECTIVE

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1 Division of Evolution, Ecology and Genetics, Research School of Biology, Australian National University, Acton, Canberra, Australia, **2** Department of Biological Sciences, Faculty of Science, Macquarie University, North Ryde, New South Wales, Australia

P-hacking

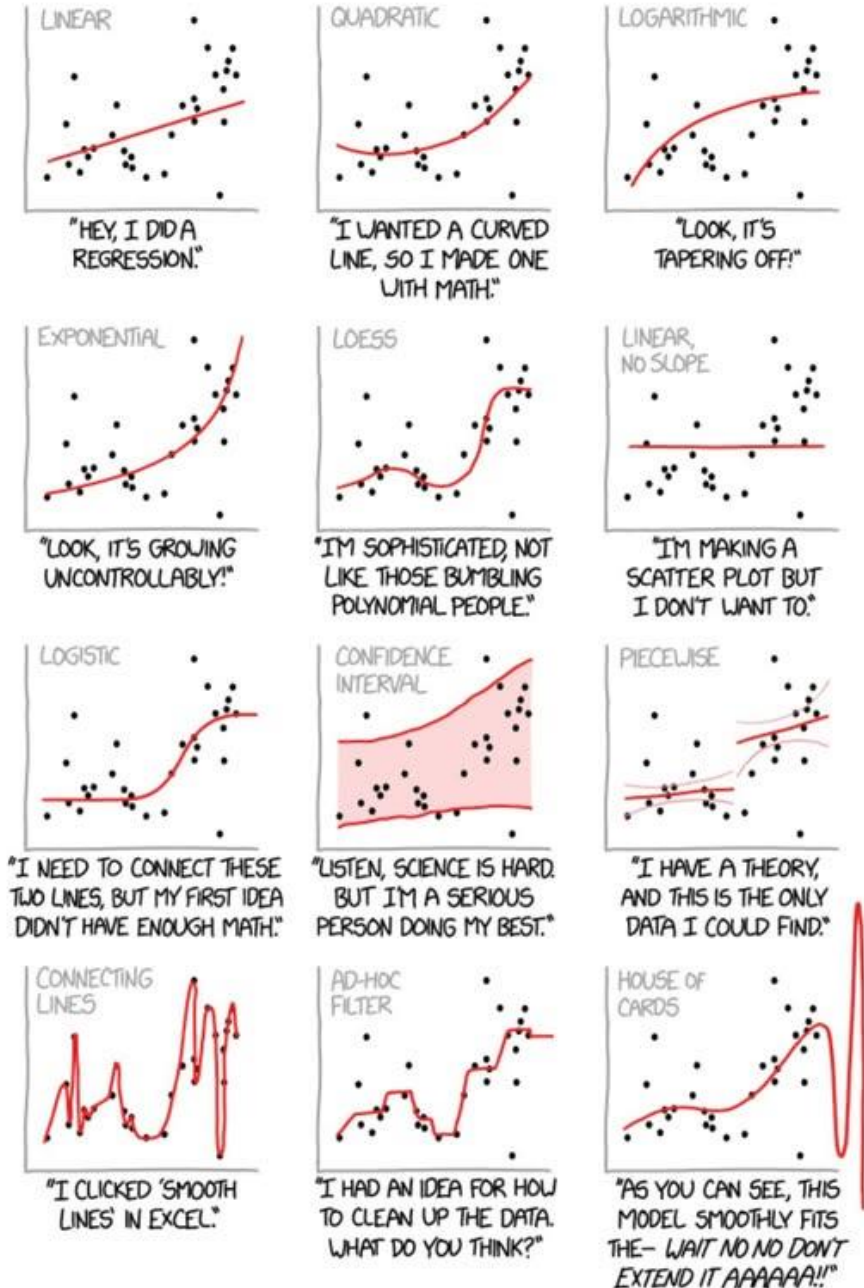
Solution?

- Pre-register hypotheses and analysis plans
- Report all analyses conducted
- Use appropriate multiple testing corrections
- Focus on effect sizes and practical significance

Overfitting

- Not really a problem with the basic methods we have learned...
... But can become one with additive models, machine learning, and all those methods that can handle lots of parameters

CURVE-FITTING METHODS AND THE MESSAGES THEY SEND



Overfitting

Solution?

- Fit models that are biologically relevant and meaningful
I.e., don't include every variable you can, but only the relevant ones
And keep shapes interpretable

Catalogue of Bias



[HOME](#) [BIASES](#) [BLOG](#) [CONTACT](#) [ABOUT](#)

Biases

Here is a list of all Biases in the Catalogue. Check back regularly as we are adding new ones over time.

Adherence bias

A distortion in outcome data that arises when participants who adhere to a study protocol or intervention differ from those who do not adhere, when that difference relates to the outcome of interest.

[Read More](#)

Admission rate bias

A distortion in assessment of the relation between an exposure and a disease that the exposure is thought to affect, which arises because the subjects studied had been admitted to hospital, typically increasing the odds ratio but in some cases reducing it. Also known as Berkson's bias.

[Read More](#)

All biases

Select a bias ▼

Search the Catalogue

search website...

Popular posts

- [Confirmation bias](#)
- [Hawthorne effect](#)
- [Selection bias](#)
- [Immortal time bias](#)

The **in mice* bot

In-mice* is a reddit bot that tracks posts in the popular [r/science](#) community. When it finds a headline that tries to frame mouse model research breakthroughs as human medical science, it adds the comment ‘...in mice*’ to the post.

For example: [The headline](#) ‘Nasal Vaccine may Aid Fight Against New Viral Variants’ may lead readers to believe that this vaccine has been tested in humans and will soon be available to the public. In reality, the [research paper](#) is about a set of mouse experiments, and it could be years before this translates to a human vaccine. A more responsible headline would be ‘Nasal Vaccine Generates Robust Antiviral Immune Response in Mouse Experiments’.

Why do we need this?

Molecular, biochemical and occasionally even behavioral research in 🐭 models can be very fruitful. But it’s important to note that many results seen in animal models will take years before they are translated to the clinic. Many don’t make it.

When scientists publish their research, most journals insist that the species used to perform research be included in the title. This helps put the research in context. But when a research paper is converted into a headline for a

Application

- Read the article below
- How much do you trust the title? Why?



[Home](#) | [US Showbiz](#) | [TV](#) | [Sports](#) | [Royals](#) | [Video](#) | [News](#) | [Crime Desk](#) | [Lifestyle](#) | [Money](#) | [U.K.](#) | [Buyline](#) | [Games](#) | [M+](#)

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Watching too much TV can KILL you: Sitting in front of the box for five hours a day increases risk of pulmonary embolism

- People who watch too much TV are at risk of suffering deadly blood clots
- More than five hours a day doubles pulmonary embolism risk, study finds
- Academics say TV viewers should stand up occasionally and drink water

Enter your search

Search



**Sign up for our free Health
newsletter**

Transform your health every Tuesday
with expert guides, real-life stories and
advice from top doctors

Take Home Message

- Don't just trust the p-value...
- Look for the data
- Look for the mechanism
- Look for the confounders
- Make sure you understand what is happening

Before Next Time (Friday April 17, in LH4)

- Due before next week
 - Check you are ready for next class: **Knowledge Check #11** due by **Friday April 17 11 AM**
- Due the follow week
 - Practice: start working on **Homework #11** due by **Thursday April 23 5 PM**
 - Use the Guiding Quiz #11 if needed

Need Help?

- Come to VTPEH 6108: Public Health Data Lab
- Come to office hours
 - **Tuesdays, 09:00-10:00 AM**
 - **Wednesdays, 05:00-06:00 PM**
 - In S1-122, unless indicated otherwise on Canvas